Exam 2 on Friday

- Bring a calculator!

ANOVA Table (G ll on review)

$\bar{y}$ = overall mean of y

$\hat{y}_i$ = prediction from model

$$Iridium = \beta_0 + \beta_1 \, ls\, Shale + \varepsilon$$

Decompose sum of squares:

$$\underbrace{\sum_{i=1}^{n} (y_i - \bar{y})^2}_{SSTotal} = \underbrace{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}_{SSE} + \underbrace{\sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2}_{SSModel}$$

Source	df	SS	MS	F
Model	$p-1$	SSModel	SSModel $/(p-1)$	$\dfrac{SSModel/(p-1)}{SSE/(n-p)}$
Residuals	$n-p$	SSE	$SSE/(n-p)$	
Total	$n-1$	SSTotal		

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- Bring a calculator!

ANOVA Table (Q11 on review)

$\bar{y}$ = overall mean of y

$\hat{y}_i$ = prediction from model

$$\text{Iridium} = \beta_0 + \beta_1 \text{ls Shale} + \varepsilon$$

Decompose sum of squares:

$$\underbrace{\sum_{i=1}^{n} (y_i - \bar{y})^2}_{\text{SSTotal}} = \underbrace{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}_{\text{SSE}} + \underbrace{\sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2}_{\text{SSModel}}$$

Source	df	SS	MS	F
model	1	88363	88363	$\frac{88363}{44768} = 1.97$
Residuals	26	1163968	$\frac{1163968}{26} = 44768$	
Total	27	1252331		

Under H_0, $F \sim F_{1,26}$

t - test vs. nested F-test

t - test

- test single parameter (i.e. one β)

- Examples:

$H_0: \beta_1 = 0$

$H_A: \beta_1 \neq 0$

here we can use either test

$t^2 = F$

$H_0: \beta_1 = 0$

$H_A: \beta_1 < 0$

$H_0: \beta_1 = 3$

$H_A: \beta_1 > 3$

nested F - test

- compare full and reduced model

- always asking whether to drop one or more parameters

$H_0: \beta_1 = 0$

$H_A: \beta_1 \neq 0$

$H_0: \beta_1 = \beta_2 = \beta_3 = 0$

H_A: at least one of $\beta_1, \beta_2, \beta_3 \neq 0$

Q 3 (review)

$t \sim t_d$ then $t^2 \sim F_{1,d}$ (Fact)

Example:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Test $H_0: \beta_1 = 0$

$H_A: \beta_1 \neq 0$

t-test: $t = \dfrac{\hat{\beta}_1 - 0}{SE\hat{\beta}_1} \sim t_{n-2}$

F-test: $F = \dfrac{SSmodel/(p-1)}{SSE/(n-p)} = \dfrac{SSmodel}{SSE/(n-p)} \sim F_{1,n-p}$

$$t^2 \equiv F$$

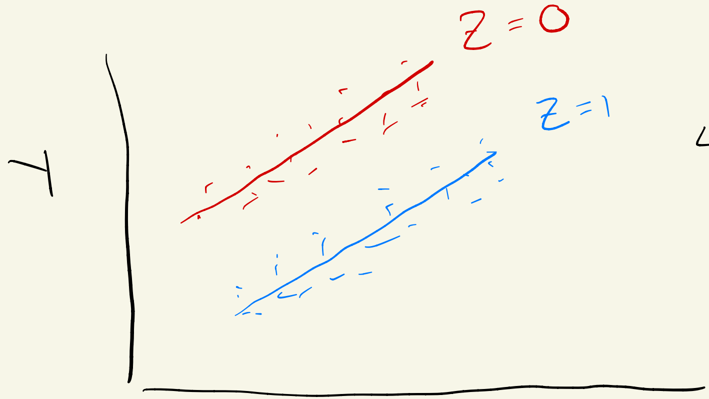
Interactions

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \varepsilon$$

This model assumes that the slope on X is the same (β_1) for every value of Z

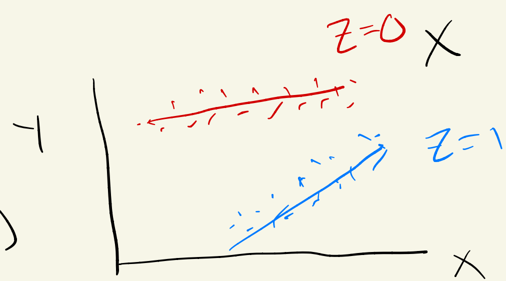
Ex:

no interaction needed



← slope on these lines is the same (parallel)

do need interaction (to allow for different slopes)



← lines are not parallel

⇒ Assumption of parallel lines is wrong

when interactions?

- If we look at our data & we see different slopes for different groups
- If our research question of interest requires us to compare slopes between groups

  ex: "Is the relationship between bill length & bill depth different for different species of penguin?"

  ex: "Is the relationship between diversity & group performance dependent on testosterone?"

Research question that doesn't necessarily require an interaction:
"Is there a positive relationship between bill length & bill depth, after accounting for species?"

R^2 = proportion of variability in response explained by regression on explanatory variables

E.g. Q21 : $R^2 = 0.5127$

51.27% of variability in final-day archery scores is explained by regression on first-day archery score & attendance

why can R^2 never decrease when we add variables?

$$R^2 = 1 - \frac{SSE}{SSTotal}$$

when we add variables, SSE can never go up

$\Rightarrow$ R^2 can never go down

$$SSE = \sum_{i=1} (y_i - \hat{y}_i)^2 = \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 X)^2$$

Add new variable : $\sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i - \hat{\beta}_2 Z_i)^2$

<u>VIF</u>:

Suppose $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$

$VIF(\hat{\beta}_1)$: Predict X_1 from X_2, X_3

$$X_1 = \alpha_0 + \alpha_1 X_2 + \alpha_2 X_3 + \varepsilon$$

$\Rightarrow R^2$

$$VIF = \frac{1}{1-R^2}$$